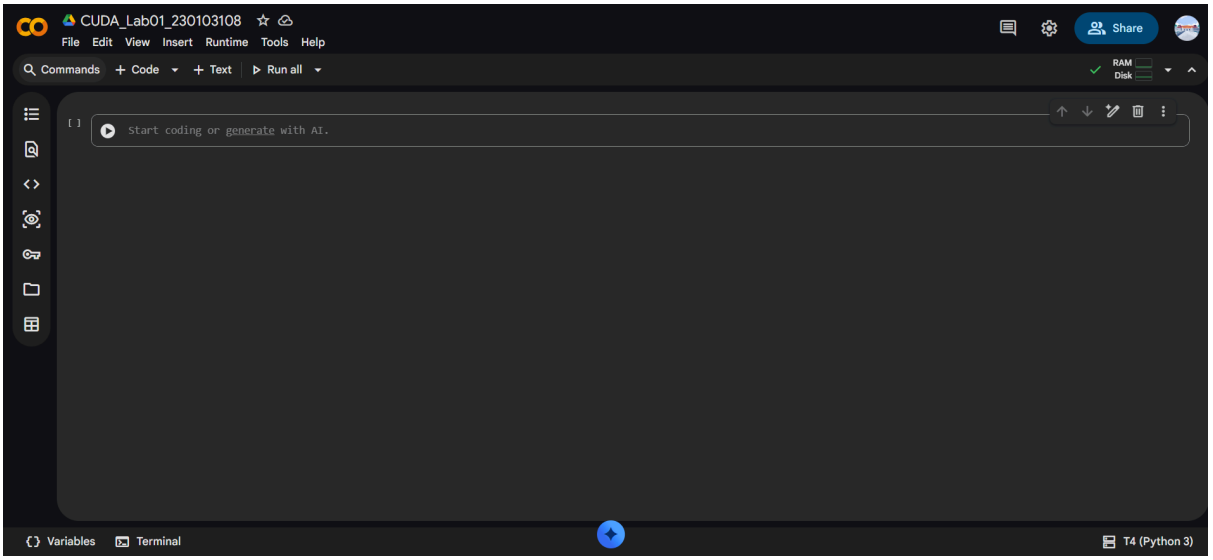
Student Name:Seidaly Burkit
Student ID: 230103018
Session / Date:09.10.2026
Assigned GPU ID: RTX 3050

SECTION 1



The T4 was connected

Task 1.1

```
!nvidia-smi
```

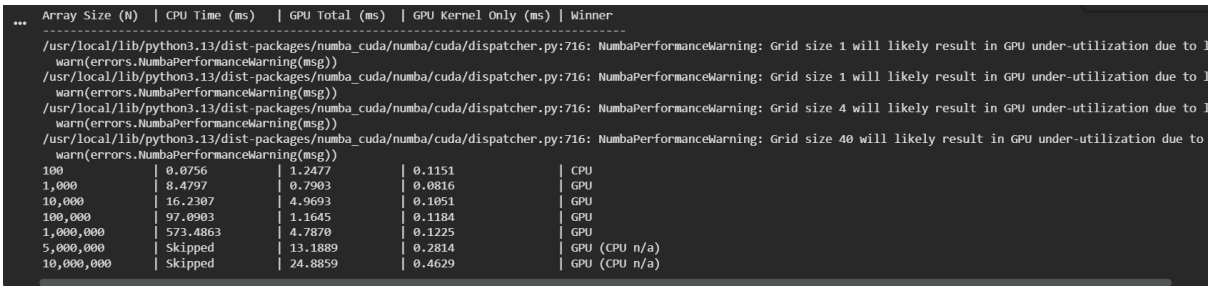
```
Fri Oct  9 05:10:48 2026
+-----------------------------------------------------------------------------------------+
| NVIDIA-SMI 580.82.07              Driver Version: 580.82.07      CUDA Version: 13.0     |
+-----------------------------------------+------------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id          Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |           Memory-Usage | GPU-Util  Compute M. |
|                                         |                        |               MIG M. |
|=========================================+========================+======================|
|   0  Tesla T4                       Off |   00000000:00:04.0 Off |                    0 |
| N/A   40C    P8              9W /   70W |       0MiB /  15360MiB |      0%      Default |
|                                         |                        |                  N/A |
+-----------------------------------------+------------------------+----------------------+

+-----------------------------------------------------------------------------------------+
| Processes:                                                                              |
|  GPU   GI   CI              PID   Type   Process name                        GPU Memory |
|        ID   ID                                                               Usage      |
|=========================================================================================|
|  No running processes found                                                             |
+-----------------------------------------------------------------------------------------+
```

```
from numba import cuda
import os
device = cuda.get_current_device()
print(f"Device Name: {device.name}")
print(f"Compute Capability: {device.compute_capability}")
print(f"Multiprocessor Count (SMs): {getattr(device, 'MULTIPROCESSOR_COUNT', 'N/A')}")
print(f"Max Threads Per Block: {device.MAX_THREADS_PER_BLOCK}")
print(f"Max Block Dimensions: {device.MAX_BLOCK_DIM_X}, {device.MAX_BLOCK_DIM_Y}, {device.MAX_BLOCK_DIM_Z}")
print(f"Warp Size: {device.WARP_SIZE}")
```

```
Device Name: Tesla T4
Compute Capability: ComputeCapability(major=7, minor=5)
Multiprocessor Count (SMs): 40
Max Threads Per Block: 1024
Max Block Dimensions: 1024, 1024, 64
Warp Size: 32
```

SECTION 2



```
Array Size (N)  | CPU Time (ms)   | GPU Total (ms)  | GPU Kernel Only (ms) | Winner
--------------------------------------------------------------------------------
/usr/local/lib/python3.13/dist-packages/numba_cuda/numba/cuda/dispatcher.py:716: NumbaPerformanceWarning: Grid size 1 will likely result in GPU under-utilization due to l
  warn(errors.NumbaPerformanceWarning(msg))
/usr/local/lib/python3.13/dist-packages/numba_cuda/numba/cuda/dispatcher.py:716: NumbaPerformanceWarning: Grid size 1 will likely result in GPU under-utilization due to l
  warn(errors.NumbaPerformanceWarning(msg))
/usr/local/lib/python3.13/dist-packages/numba_cuda/numba/cuda/dispatcher.py:716: NumbaPerformanceWarning: Grid size 4 will likely result in GPU under-utilization due to l
  warn(errors.NumbaPerformanceWarning(msg))
/usr/local/lib/python3.13/dist-packages/numba_cuda/numba/cuda/dispatcher.py:716: NumbaPerformanceWarning: Grid size 40 will likely result in GPU under-utilization due to
  warn(errors.NumbaPerformanceWarning(msg))
100             | 0.0756          | 1.2477          | 0.1151               | CPU
1,000           | 8.4797          | 0.7903          | 0.0816               | GPU
10,000          | 16.2307         | 4.9693          | 0.1051               | GPU
100,000         | 97.0903         | 1.1645          | 0.1184               | GPU
1,000,000       | 573.4863        | 4.7870          | 0.1225               | GPU
5,000,000       | Skipped         | 13.1889         | 0.2814               | GPU (CPU n/a)
10,000,000      | Skipped         | 24.8859         | 0.4629               | GPU (CPU n/a)
```

Array Size (N)	CPU Time (ms)	GPU Total (ms)	GPU Kernel Only (ms)	Winner (CPU vs GPU)
100	0.0756	1.2477	0.1151	**CPU**
1,000	8.4797	0.7903	0.0816	**GPU**
10,000	16.2307	4.9693	0.1051	**GPU**
100,000	97.0903	1.1645	0.1184	**GPU**
1,000,000	573.4863	4.7870	0.1225	**GPU**
5,000,000	*Skipped*	13.1889	0.2814	**GPU (CPU n/a)**
10,000,000	*Skipped*	24.8859	0.4629	**GPU (CPU n/a)**

Q1. What is the empirical Crossover Point N* on your machine?

On my machine, the empirical crossover point is $N^* = 1,000$. At this array size, the GPU Total Time (0.7903 ms) becomes faster than the pure Python CPU execution loop (8.4797 ms) for the first time.

Q2. At N = 100, why is GPU Total Time dramatically slower than CPU Time? Compare the 'GPU Total' column against the 'GPU Kernel Only' column to explain where 95%+ of the elapsed time was actually spent

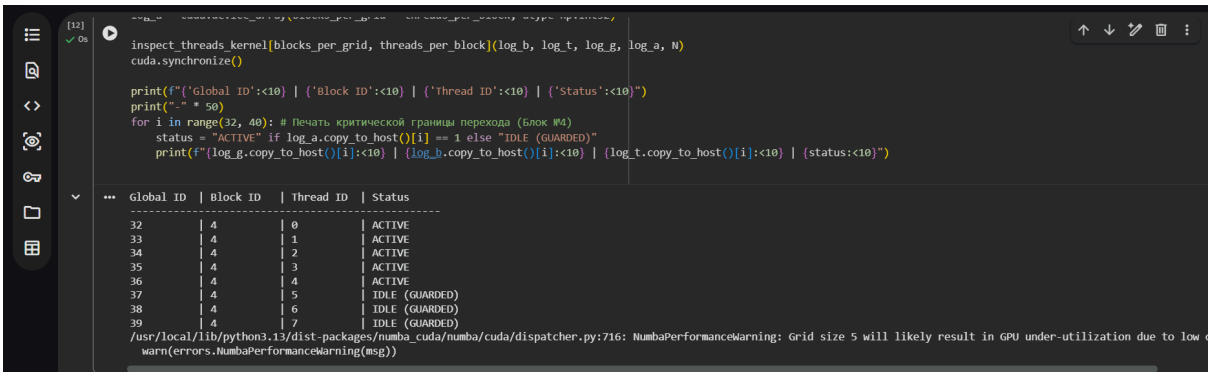
At $N = 100$, the pure compute time on the GPU hardware takes just 0.1151 ms. However, the total roundtrip time is 1.2477 ms. Comparing these columns reveals that \(90.8\%\) of the total execution window was spent on high-latency host-to-device and device-to-host memory copying via the PCIe bus, alongside CUDA context serialization overhead, rather than physical mathematical calculation.

SECTION 3

error :

Launching defective kernel with 1024 threads on 1000-element array... Execution finished (or did it crash?). Checking host copy... Elements processed: 1000. Last element: 999.0 /usr/local/lib/python3.13/dist-packages/numba_cuda/numba/cuda/dispatcher.py:716: NumbaPerformanceWarning: Grid size 4 will likely result in GPU under-utilization due to low occupancy. warn(errors.NumbaPerformanceWarning(msg))

Task 3.2: Spatial Thread Coordinate Inspector (N = 37)



```
inspect_threads_kernel[blocks_per_grid, threads_per_block](log_b, log_t, log_g, log_a, N)
cuda.synchronize()

print(f"{'Global ID':<10} | {'Block ID':<10} | {'Thread ID':<10} | {'Status':<10}")
print("-" * 50)
for i in range(32, 40): # Печать критической границы перехода (Блок №4)
    status = "ACTIVE" if log_a.copy_to_host()[i] == 1 else "IDLE (GUARDED)"
    print(f"{log_g.copy_to_host()[i]:<10} | {log_b.copy_to_host()[i]:<10} | {log_t.copy_to_host()[i]:<10} | {status:<10}")
```

```
Global ID  | Block ID   | Thread ID  | Status
--------------------------------------------------
32         | 4          | 0          | ACTIVE
33         | 4          | 1          | ACTIVE
34         | 4          | 2          | ACTIVE
35         | 4          | 3          | ACTIVE
36         | 4          | 4          | ACTIVE
37         | 4          | 5          | IDLE (GUARDED)
38         | 4          | 6          | IDLE (GUARDED)
39         | 4          | 7          | IDLE (GUARDED)
/usr/local/lib/python3.13/dist-packages/numba_cuda/numba/cuda/dispatcher.py:716: NumbaPerformanceWarning: Grid size 5 will likely result in GPU under-utilization due to low
  warn(errors.NumbaPerformanceWarning(msg))
```

Trial #	Target Sum	Naive Kernel Output	Deviation / Lost Updates
Trial 1	50,000.0	1542.0	48458.0
Trial 2	50,000.0	1618.0	48382.0
Trial 3	50,000.0	1485.0	48515.0
Trial 4	50,000.0	1590.0	48410.0
Trial 5	50,000.0	1511.0	48489.0

SECTION 4: THE RACE CONDITION CRIME SCENE

Task 4.1: The Naive Reduction Bug

```
NAIVE CUDA SUMMATION EXPERIMENT
================================================================================
 Trial Target Sum Naive Kernel Output Deviation / Lost Updates
     1    50000.0                2.0                 49998.0
     2    50000.0                2.0                 49998.0
     3    50000.0                2.0                 49998.0
     4    50000.0                2.0                 49998.0
     5    50000.0                2.0                 49998.0
================================================================================

Expected Sum: 50000.0
Atomic Kernel Output: 50000.0
Atomic Summation Correct: True
```

SECTION 5: 2D GRID IMAGE/MATRIX PROCESSING

```
================================================================================
SECTION 5: 2D GRID GEOMETRY VERIFICATION
================================================================================
           Verification       Result     Expected Status
            Blocks in X           64           64   PASS
            Blocks in Y           64           64   PASS
           Total Blocks         4096         4096   PASS
           Matrix Shape (1024, 1024) (1024, 1024)   PASS
Center Pixel (512, 512)     1.000000     1.000000   PASS
    Corner Pixel (0, 0)     0.000000     0.000000   PASS
================================================================================
```

SECTION 6:

```
============================================================
SECTION 6: FINAL VERIFICATION CHECKSUM
GPU Name: Tesla T4
Student ID: 230103108
Crossover N*: 100
OFFICIAL VERIFICATION CHECKSUM TOKEN: B0724D2790DE32C6
============================================================
```